SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Data Interpretation

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO3 –Data Interpretation**	**Assessment:**	**Assessment Tool 2– Data Interpretation**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Jothika M

Criteria	**Weigh t age**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**	**Mark s**
Understandin g of Case	**25**	**Thorough understanding ; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding ; some issues missed**	**Poor understanding ; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

SIMATS ENGINEERING

COURSE CODE: DSA0405 - FUNDAMENTALS OF DATA SCIENCE

CO 3 - ASSESSMENT TOOL 2: DATA INTERPRETATION SOLVED ANSWER KEY (QUESTIONS 1–5)

Jothika M (192324235)

Question 1 — Data Summarization

A dataset of 11 employee salaries (in ₹'000) is given as: 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120.

(a) Compute the Mean and Median

Step 1: Arrange the data (already sorted)

32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120 (n = 11)

Step 2: Compute the Mean

$$\text{Mean } (\bar{x}) = (\Sigma x) / n$$

Sum of all values = 32+35+36+38+40+42+45+48+50+95+120 = 581

$$\text{Mean} = 581 / 11 = 52.82 \text{ (approximately ₹52,820)}$$

Step 3: Compute the Median

Since n = 11 is odd, the median is the value at position (n+1)/2 = 6th position in the sorted array.

$$\text{Median} = \text{6th value} = 42 \text{ (₹42,000)}$$

(b) Why the Mean Is a Poor Measure of Central Tendency Here

The mean (₹52,820) is noticeably higher than the median (₹42,000) and higher than nine out of the eleven observed salaries. This happens because the mean is calculated by summing every value and dividing by the count, which means every observation — including extreme ones — contributes proportionally to the total. The two largest salaries, ₹95,000 and ₹120,000, are far removed from the bulk of the data (which clusters between ₹32,000 and ₹50,000) and therefore pull the arithmetic mean strongly upward.

The median, in contrast, depends only on the rank/position of values and is unaffected by how extreme the outlying values are — whether the top salary were ₹120,000 or ₹1,200,000, the median would remain 42. Because the mean is sensitive to these two unusually high salaries while the median is not, the mean gives a distorted impression of the 'typical' employee salary. A prospective employee reading only the mean might believe a 'typical' salary is around ₹52,800, when in fact 9 of the 11 employees (about 82%) earn ₹50,000 or less. This is a textbook illustration of why the mean is described as sensitive to outliers and skewed distributions, while the median is described as a robust measure of central tendency.

(c) Standard Deviation vs. IQR — Which Measure of Dispersion to Trust

For this dataset, the Interquartile Range (IQR) should be trusted more than the standard deviation as a measure of spread.

- Standard deviation uses squared deviations from the mean, so extreme values are weighted even more heavily than in the mean itself (because the deviation is squared). The two high salaries (95 and 120) inflate the standard deviation substantially, making it a measure that reflects the spread of a distorted center rather than the spread of the 'typical' salary band.
- IQR (Q3 − Q1) measures the spread of only the middle 50% of the data, and is completely unaffected by how far the top or bottom 25% of values extend. It therefore reflects the dispersion of the bulk of ordinary salaries without being distorted by the two extreme outliers.

Because this dataset is clearly right-skewed with two strong high-value outliers, the IQR is the more trustworthy and interpretable measure of dispersion, exactly as the median is the more trustworthy measure of central tendency in the same situation. Standard deviation would be the preferred choice only if the data were approximately symmetric and free of significant outliers.

Supporting Calculation: Standard Deviation of the Dataset

To make the comparison concrete, the sample standard deviation of the full dataset is calculated below using the standard formula for a sample:

$$SD = \sqrt{[\ \Sigma(x - \bar{x})^2 / (n - 1)\ ]}$$

x	$x - \bar{x}$ ($\bar{x}$=52.82)	$(x - \bar{x})^2$
32	-20.82	433.5
35	-17.82	317.5
36	-16.82	282.9
38	-14.82	219.6
40	-12.82	164.4
42	-10.82	117.1
45	-7.82	61.2
48	-4.82	23.2
50	-2.82	7.9
95	42.18	1779.1
120	67.18	4513.2

Summing the squared deviations gives approximately 7,919.6. Dividing by (n − 1) = 10 gives a sample variance of about 791.96, and taking the square root gives a sample standard deviation of approximately ₹28,140 (SD ≈ 28.14 in thousands). Notice how disproportionately large the squared deviations for 95 and 120 are (1,779.1 and 4,513.2) compared to every other value — together they account for roughly 79% of the total sum of squared deviations, even though they represent only 2 of the 11 observations (about 18% of the data). This numerically confirms why the standard deviation is so heavily distorted by these two outliers and reinforces the recommendation to rely on the IQR instead.

Question 2 — Outlier Treatment

Using the same dataset — 32, 35, 36, 38, 40, 42, 45, 48, 50, 95, 120 — apply the 1.5×IQR rule to identify outliers.

Step 1: Locate the Median and Split the Data

With n = 11, the median (6th value) is 42. Excluding the median itself, the data splits into a lower half and an upper half of 5 values each:

- Lower half: 32, 35, 36, 38, 40
- Upper half: 45, 48, 50, 95, 120

Step 2: Compute Q1 and Q3

$$Q1 = \text{median of lower half} = 36$$

$$Q3 = \text{median of upper half} = 50$$

Step 3: Compute the IQR

$$IQR = Q3 - Q1 = 50 - 36 = 14$$

Step 4: Compute the Lower and Upper Fences

$$\text{Lower Fence} = Q1 - 1.5 \times IQR = 36 - (1.5 \times 14) = 36 - 21 = 15$$

$$\text{Upper Fence} = Q3 + 1.5 \times IQR = 50 + (1.5 \times 14) = 50 + 21 = 71$$

Step 5: Identify Outliers

Any value below 15 or above 71 is flagged as a statistical outlier. Checking each observation against the fences:

Value (₹'000)	Below 15?	Above 71?	Outlier?
32	No	No	No
35	No	No	No
36	No	No	No
38	No	No	No

40	No	No	No
42	No	No	No
45	No	No	No
48	No	No	No
50	No	No	No
95	No	Yes	Outlier
120	No	Yes	Outlier

Two values — ₹95,000 and ₹120,000 — lie above the upper fence of ₹71,000 and therefore qualify as outliers under the 1.5×IQR rule. No values fall below the lower fence of ₹15,000, so there are no low-end outliers.

Step 6: Effect of Winsorization on the Mean

Winsorization replaces (caps) outlier values with the nearest fence value rather than deleting them. Applying this here, both 95 and 120 are capped at the upper fence of 71:

Winsorized dataset: 32, 35, 36, 38, 40, 42, 45, 48, 50, 71, 71

$$New\ Sum = 581 - 95 - 120 + 71 + 71 = 508$$

$$New\ Mean = 508 / 11 \approx 46.18\ (₹46{,}180)$$

Qualitatively, capping the two extreme salaries pulls the mean down from ₹52,820 to approximately ₹46,180 — a noticeably smaller value that sits much closer to the median (₹42,000) and to the bulk of the data. Winsorization reduces the influence of extreme values without discarding data points entirely (unlike trimming/removal, which would reduce the sample size), and it makes the mean a more representative summary of central tendency while still acknowledging that some employees earn more than the 'typical' range.

Winsorization vs. Outlier Removal — A Practical Comparison

Approach	New Mean	Sample Size	Effect on Distribution
Original data	52.82	11	Strong right skew, mean pulled far above median
Winsorized (cap at 71)	46.18	11	Reduced skew; extreme influence limited but retained

Outliers removed entirely	40.67	9	Mean closest to median but sample size reduced by ~18%

(For reference: removing 95 and 120 entirely leaves a 9-value dataset summing to 366, giving a mean of 366/9 ≈ 40.67.) Winsorization is generally preferred over outright removal in situations such as payroll analytics, where discarding an employee's record entirely could bias downstream analysis (e.g., headcount-based reporting) even though the raw salary value is extreme. Capping preserves the sample size and acknowledges that the extreme observations are real, legitimate data points, while still preventing them from dominating summary statistics such as the mean.

Question 3 — Measuring Asymmetry

Two datasets share the same mean (50) and standard deviation (10). Dataset A has a Pearson skewness coefficient of +1.2, and Dataset B has a coefficient of −0.8.

Interpreting the Skewness Coefficients

The sign and magnitude of Pearson's coefficient of skewness describe both the direction and the degree of asymmetry of a distribution relative to a perfectly symmetric (zero-skew) bell curve:

- A positive coefficient (Dataset A, +1.2) indicates positive (right) skew: the bulk of observations cluster at the lower end of the scale, with a long tail extending toward higher values. The mean is pulled above the median, which is in turn above the mode (Mean > Median > Mode).
- A negative coefficient (Dataset B, −0.8) indicates negative (left) skew: the bulk of observations cluster at the higher end of the scale, with a long tail extending toward lower values. The mean is pulled below the median, which is below the mode (Mean < Median < Mode).

Shape Description

Dataset A (Skewness = +1.2)

Picture a histogram with a tall peak positioned to the left of center (around values somewhat below 50), a short, quickly rising left side, and a long, gradually thinning tail stretching out to the right toward much higher values (well beyond 70–80). Because the magnitude of skew (1.2) is fairly large, this asymmetry is strongly visible rather than subtle — the distribution looks noticeably lopsided, not just mildly tilted.

Dataset B (Skewness = −0.8)

Picture a histogram with a peak positioned to the right of center (around values somewhat above 50), a short, steep right side, and a moderately long tail stretching out to the left toward lower values. The magnitude (0.8) is smaller than Dataset A's, so the asymmetry is present and noticeable but less extreme than in Dataset A.

Which Dataset Resembles Salary-Like Data?

Dataset A, with its strong positive skewness (+1.2) and long right tail, is far more likely to represent salary-like data. Real-world salary distributions are a classic example of positively skewed data: the majority of employees earn salaries clustered in a moderate range, while a small number of executives, senior specialists, or top performers earn disproportionately large salaries that stretch the distribution's tail far to the right — exactly the pattern seen with the two high earners (₹95,000 and ₹120,000) in Question 1's dataset. Dataset B's negative skew, with a long left tail, would instead be typical of phenomena where most values are high and only a few fall far below the norm (for example, exam scores on an easy test, or age at retirement in a population where most people retire around a fixed age but a few retire much earlier).

Real-World Examples of Each Skew Direction

Skew Type	Typical Real-World Examples	Mean vs Median
Positive (right) skew — like Dataset A	Salaries/income, house prices, hospital length-of-stay, insurance claim amounts	Mean > Median > Mode
Negative (left) skew — like Dataset B	Age at retirement, exam scores on an easy test, time-to-failure of long-lasting products	Mean < Median < Mode
No skew (symmetric)	Heights of adults within one gender, standardized IQ scores, measurement error	Mean ≈ Median ≈ Mode

This pattern — Mean greater than Median for positive skew, and Mean less than Median for negative skew — is a quick diagnostic that analysts routinely use before even computing a formal skewness coefficient. Simply comparing the mean and median of the salary dataset in Question 1 (52.82 vs 42) immediately signals strong positive skew, consistent with the shape described for Dataset A here.

Question 4 — Coefficient of Skewness Calculation

Given: Mean = 62, Median = 58, Mode = 52, Standard Deviation = 12.

(a) Karl Pearson's Coefficient — Mode Method

Sk = (Mean − Mode) / SD

Sk = (62 − 52) / 12 = 10 / 12 = 0.833

Using the mode-based formula, the coefficient of skewness is approximately +0.83, indicating a moderate positive (right) skew.

(b) Karl Pearson's Coefficient — Median Method

Sk = 3 × (Mean − Median) / SD

Sk = 3 × (62 − 58) / 12 = 3 × 4 / 12 = 12 / 12 = 1.0

Using the median-based (empirical) formula, the coefficient of skewness is exactly +1.0, indicating a somewhat stronger positive skew than suggested by the mode-based method.

Comparison and Explanation of the Difference

Method	Formula	Result	Interpretation
Mode method	(Mean − Mode) / SD	0.833	Moderate positive skew
Median method	3(Mean − Median) / SD	1.000	Slightly stronger positive skew

Both formulas agree that the distribution is positively skewed (Mean > Median > Mode, consistent with a right-skewed shape), but they produce slightly different numerical values — 0.833 versus 1.0. This difference arises because the median-based formula is not an independent measure; it is a mathematical approximation derived from the empirical relationship between the three measures of central tendency in a moderately skewed unimodal distribution:

Empirical relation: Mode ≈ 3 × Median − 2 × Mean

If the actual mode exactly satisfied this empirical relationship, both formulas would yield identical results. Substituting the given mean and median into the empirical relation predicts a mode of 3(58) − 2(62) = 174 − 124 = 50, but the actual/observed mode given in the problem is 52 — a small discrepancy of 2 units. Because the mode-method formula uses the true (observed) mode of 52 rather than the empirically predicted mode of 50, its result (0.833) differs slightly from the median-method result (1.0), which relies purely on the empirical approximation and does not use the mode at all. In practice, when the observed mode closely matches the empirically predicted mode, the two skewness coefficients will be nearly identical; larger gaps between the observed and predicted mode (as seen here) lead to more noticeable differences between the two coefficients. The median-based formula is generally preferred in practice because the mode is often difficult to determine precisely or may be ill-defined (e.g., in continuous or multimodal data), whereas the mean and median can always be computed unambiguously.

Cross-Check Using Bowley's Coefficient of Skewness (Conceptual Note)

A third, quartile-based approach — Bowley's coefficient of skewness — is often used as an additional cross-check when quartile data (Q1, Q3) is available:

$$\text{Bowley's } Sk = (Q3 + Q1 - 2\times\text{Median}) / (Q3 - Q1)$$

Bowley's coefficient is not directly computable here because Q1 and Q3 were not provided for this particular dataset (only Mean, Median, Mode and SD are given). It is mentioned for completeness: like the Pearson mode- and median-based coefficients, Bowley's measure would also be expected to return a positive value for this distribution, since all indicators (Mean > Median > Mode) consistently point to positive skewness. Having multiple, largely consistent skewness estimates from different formulas increases confidence that the distribution is indeed moderately-to-strongly positively skewed, even though the exact numerical coefficient varies slightly (0.833 vs 1.0) depending on which central-tendency measures the formula relies upon.

Question 5 — Variance vs. Standard Deviation Interpretation

Two production lines report the following statistics:

Line	Mean (units/day)	Variance
Line A	200	64
Line B	150	36

Step 1: Why Raw Variance Is Misleading Here

At first glance, Line A's variance (64) is nearly double Line B's variance (36), which might tempt one to conclude that Line A is 'more variable.' However, variance is expressed in squared units and, critically, its magnitude is heavily influenced by the scale of the mean. Line A operates at a higher average output (200 units/day) than Line B (150 units/day), so comparing their variances directly is like comparing apples grown on two differently sized trees — a larger mean naturally tends to be associated with a larger absolute spread even if the relative (proportional) spread is the same. Raw variance (or standard deviation) is therefore not the right tool for comparing variability across datasets with different means or units; the Coefficient of Variation (CV) is designed for exactly this purpose.

Step 2: Compute Standard Deviations

$SD(Line\ A) = \sqrt{Variance} = \sqrt{64} = 8\ units/day$

$SD(Line\ B) = \sqrt{Variance} = \sqrt{36} = 6\ units/day$

Step 3: Compute the Coefficient of Variation

$CV = (Standard\ Deviation\ /\ Mean) \times 100\%$

$CV(Line\ A) = (8\ /\ 200) \times 100 = 4.0\%$

$CV(Line\ B) = (6\ /\ 150) \times 100 = 4.0\%$

Step 4: Interpretation

Metric	Line A	Line B	Conclusion
Standard Deviation	8 units/day	6 units/day	A appears more variable in absolute terms
Coefficient of Variation	4.0%	4.0%	Equal relative variability

Once the spread is expressed relative to each line's own average output, both production lines show exactly the same relative variability: 4.0%. This means that, proportionally, Line A and Line B fluctuate around their respective means to the same degree — Line A simply operates at a higher absolute output level, so the same relative fluctuation translates into a larger absolute variance and standard deviation. If a plant manager judged consistency using variance alone, they would incorrectly conclude Line A is less consistent; using the Coefficient of Variation instead reveals that both lines are equally consistent relative to their own production scale. This demonstrates why CV — not raw variance or standard deviation — is the appropriate statistic for comparing variability between datasets that have different means, different units, or different scales of measurement.

Key Takeaway

Coefficient of Variation is a dimensionless, scale-independent measure of relative dispersion, making it uniquely suited to answering questions like 'which process/line/dataset is relatively more variable?' whenever the datasets being compared do not share the same mean. Raw variance and standard deviation remain useful for describing absolute spread within a single dataset or between datasets with equal or similar means, but they should not be used alone to rank variability across datasets operating at different scales.

When to Use CV vs. Raw Variance/SD — General Guidance

Situation	Recommended Measure	Reason
Comparing two datasets with similar means/units	Variance or SD	Absolute spread is directly comparable
Comparing datasets with different means or units	Coefficient of Variation	CV normalizes spread relative to scale
Comparing consistency of two machines with different output levels	Coefficient of Variation	Removes bias from differing average output
Ranking risk of financial assets with different average returns	Coefficient of Variation	Standard practice in finance (risk per unit return)
Quality control within a single production line over time	Standard Deviation / Control Charts	Mean is constant, so absolute spread is meaningful

This case also illustrates an important general principle in data interpretation: a statistic that appears larger in absolute terms does not necessarily indicate 'more variability' in any meaningful sense once differences in scale are accounted for. Analysts should always ask whether the quantities being compared share a common baseline before drawing conclusions from raw dispersion measures alone.

Glossary of Statistical Terms Used

Term	Definition
Mean	The arithmetic average of all values in a dataset; sum of values divided by count.
Median	The middle value of a sorted dataset; the 50th percentile.
Mode	The most frequently occurring value in a dataset.
Variance	The average of the squared deviations of each value from the mean; measures spread in squared units.
Standard Deviation	The square root of variance; measures spread in the same units as the original data.
Quartile (Q1, Q3)	Values that divide a sorted dataset into quarters; Q1 = 25th percentile, Q3 = 75th percentile.

Interquartile Range (IQR)	The difference Q3 − Q1; measures the spread of the middle 50% of data, robust to outliers.
Outlier	An observation that lies unusually far from the rest of the data, often defined using the 1.5×IQR rule.
Winsorization	A technique that caps extreme values at a specified fence rather than removing them.
Skewness	A measure of the asymmetry of a probability distribution around its mean.
Coefficient of Variation (CV)	The ratio of standard deviation to mean, expressed as a percentage; a scale-free measure of relative variability.

Appendix: Formula Reference Sheet

Central Tendency

Mean ($\bar{x}$) = $\Sigma x / n$

Median = middle value of sorted data (or average of two middle values if n is even)

Mode = most frequently occurring value

Dispersion

Variance (sample) = $\Sigma(x - \bar{x})^2 / (n - 1)$

Standard Deviation = $\sqrt{\text{Variance}}$

Range = Maximum − Minimum

IQR = Q3 − Q1

Coefficient of Variation (CV) = (SD / Mean) × 100%

Outlier Detection

Lower Fence = Q1 − 1.5 × IQR

Upper Fence = Q3 + 1.5 × IQR

Skewness

Karl Pearson's Sk (mode method) = (Mean − Mode) / SD

Karl Pearson's Sk (median method) = 3 × (Mean − Median) / SD

Bowley's Sk = (Q3 + Q1 − 2×Median) / (Q3 − Q1)

Empirical relation = Mode ≈ 3×Median − 2×Mean

Rule-of-Thumb Interpretation Guide

Skewness Value	Interpretation
Sk = 0	Perfectly symmetric distribution
0 < Sk < 0.5 (or −0.5 < Sk < 0)	Approximately symmetric / fairly symmetrical
0.5 ≤ Sk < 1 (or −1 < Sk ≤ −0.5)	Moderately skewed
Sk ≥ 1 (or Sk ≤ −1)	Highly / strongly skewed

Summary of Key Results

Q.No	Concept Tested	Key Result
1	Mean, Median, SD vs IQR	Mean = 52.82, Median = 42; IQR preferred over SD
2	1.5×IQR Outlier Rule	Q1=36, Q3=50, IQR=14; Outliers = 95, 120; Winsorized mean ≈ 46.18
3	Skewness Sign & Shape	Dataset A (+1.2): right-skewed, salary-like; Dataset B (−0.8): left-skewed
4	Karl Pearson's Skewness	Mode method = 0.833; Median method = 1.0
5	Variance, SD & Coefficient of Variation	CV(A) = CV(B) = 4.0% — equal relative variability

Frequently Confused Concepts — Quick Clarifications

- Variance vs. Standard Deviation: Variance is in squared units (e.g., 'units²'), which makes it hard to interpret directly; standard deviation converts this back into the original units, making it more intuitive for reporting.
- IQR vs. Range: Range uses only the two extreme values (max − min) and is highly sensitive to outliers; IQR uses the middle 50% of data and is robust to outliers.
- Skewness vs. Kurtosis: Skewness measures asymmetry (lopsidedness) of a distribution; kurtosis measures 'tailedness' or peakedness, a related but distinct property not covered in these questions.

- Population vs. Sample Formulas: Variance/SD formulas divide by n for a population but by (n − 1) for a sample (Bessel's correction), since a sample tends to slightly underestimate true population variance otherwise.
- Outlier vs. Extreme Value: Not every large or small value is a statistical outlier — the 1.5×IQR rule (or similar formal criteria) should be used to make this determination objectively rather than by visual judgement alone.

Concluding Remarks

Across all five questions, a consistent theme emerges: summary statistics can only be interpreted correctly when their sensitivity to outliers, scale, and distributional shape is properly understood. The mean, variance, and standard deviation are powerful and widely used, but they are all sensitive to extreme values and to the absolute scale of the data. The median, IQR, coefficient of variation, and skewness coefficients provide the necessary context to avoid drawing misleading conclusions — whether that means recognizing that a 'typical' salary is better represented by the median than the mean, correctly flagging and treating outliers before further analysis, understanding the direction and strength of asymmetry in a distribution, reconciling small differences between alternative skewness formulas, or fairly comparing variability between datasets that operate on different scales. Mastery of these distinctions is central to sound exploratory data analysis and to making reliable, data-driven decisions.